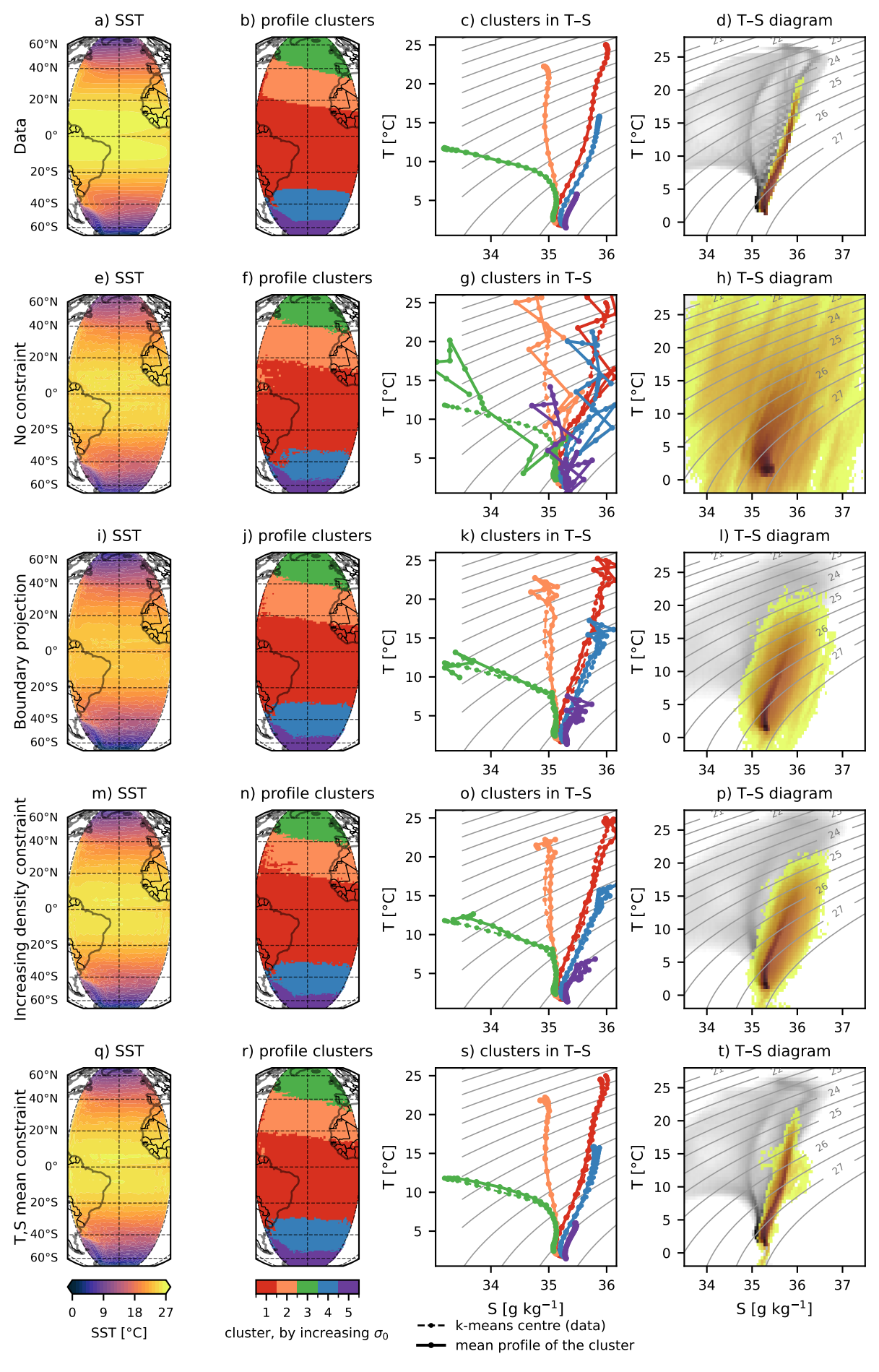

a) SST
b) profile clusters
c) clusters in T–S
d) T–S diagram
Data
e) SST
f) profile clusters
g) clusters in T–S
h) T–S diagram
No constraint
i) SST
j) profile clusters
k) clusters in T–S
l) T–S diagram
Boundary projection
m) SST
n) profile clusters
o) clusters in T–S
p) T–S diagram
Increasing density constraint
q) SST
r) profile clusters
s) clusters in T–S
t) T–S diagram
T,S mean constraint
60°N
40°N
20°N
0°
20°S
40°S
60°S
T [°C]
S [g kg^{-1}]
0 9 18 27
SST [°C]
1 2 3 4 5
cluster, by increasing σ_0
k-means centre (data)
mean profile of the cluster